

RENANCE

UNIVERSITY OF PORT HARCOURT

MTH 101 — Past Questions & Materials

by Renance

Course: MTH 101

Institution: University of Port Harcourt

Papers in this pack: 1

Levels: 100

This study pack was compiled and restructured by Renance for MTH 101 students at University of Port Harcourt. It brings together 1 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

Study online: renance.app

Get the app: Google Play — [dev.renance.renance](https://play.google.com/store/apps/details?id=dev.renance.renance)

Generated by Renance AI

Paper 1 of 1 — Elementary set theory and algebra MTH 101

Level 100 • First Semester • Handout

This source paper is a scanned document; its pages are reproduced below.

UPLOADED BY
 ABU JAWAL

UNIVERSITY OF ABUJA
 FACULTY OF SCIENCE
 DEPARTMENT OF MATHEMATICS
 2016/2017 FIRST SEMESTER EXAMINATION
 MTH 101: ELEMENTARY SET THEORY AND ALGEBRA UNITS: 2 TIME: 2HOURS
 TYPE A

INSTRUCTION: Answer all questions in section (A) and one question in section (B)
 Use H.B. pencil to shade the OMR Sheet and to write the question type
 SECTION A

Use the information below to answer questions 1 to 3

All the 1000 students in Faculty of Science, University of Abuja were asked their subject combinations. 360 of them offered Mathematics, 420 offered Chemistry while 320 offered Biology. 140 students offered Mathematics and Chemistry, 160 offered Mathematics and Biology, 180 students offered Chemistry and Biology while 100 offered the three subject combinations. Find

- The number of students that offered Mathematics but offered neither Chemistry nor Biology
 (a) 180 (b) 100 (c) 160 (d) 140
- The number of students who did not offer any of the three subject combinations
 (a) 720 (b) 440 (c) 420 (d) 380
- The number of students who offer at least two subjects. (a) 280 (b) 360 (c) 420 (d) 380

Use the information below to answer questions 4 to 6

Let $U = \{\text{all natural numbers less than or equal to } 20\}$.
 $A = \{\text{factors of } 12\}$, $B = \{x: 2 \leq x \leq 10; x \text{ is even number}\}$.
 $C = \{\text{factors of } 16\}$

List the members of:

- $A' \cap B' \cap C$.
 (a) $\emptyset$ (b) $\{1, 2, 4, 8, 16\}$ (c) $\{2, 4, 6, 8, 10\}$ (d) $\{7, 9, 11, 13, 14, 15, 16, 17, 18, 19, 20\}$
- $A' \cup B \cap C'$.
 (a) $\{3, 5, 7, \dots, 20\}$ (b) $\{3, 5, 6, 7, 9, 11, 13, \dots, 20\}$ (c) $\{3, 5, 6, 7, 9, 11, 13, 14, 15, 17, \dots, 20\}$ (d) U
- $(A \cup B \cup C)'$
 (a) $\{3, 5, 7, 9\}$ (b) $\{5, 7, 9, 11, 13, 14, 15, 17, 18, 19, 20\}$ (c) $\{7, 9, 11, \dots, 20\}$ (d) U

7. A binary operation $*$ is defined on the set R of real numbers by;
 $a * b = a + b + ab$. Find the identity element e of R under the operation
 (a) a (b) 1 (c) $1 - a$ (d) 0

8. The operation $*$ on the set R of real numbers is defined by
 $a * b = \frac{2a-1}{2} + b \quad \forall a, b \in R$. Find the inverse of element $x \in R$
 (a) $1 - a$ (b) $a - 1$ (c) $2 - 2a$ (d) $1 - 2a$

9. Determine the domain D of the mapping $f: x \rightarrow 2x + 3$, if $C = \{-3, -1, 5\}$ is the range and f is defined on D .
 (a) $\{1, 2, 3\}$ (b) $\{-5, -3, 1\}$ (c) $\{-3, -2, 1\}$ (d) $\{-3, 1, 7\}$

10. If f and g are mappings defined on the set of real numbers by $f(x) = 3x + 1$, $g(x) = x^2$. Find $f \circ g(x)$. (a) $(3x + 1)^2$ (b) $3x^2 + 1$ (c) $9x^2 + 6x + 1$ (d) $x^2(3x + 1)$

11. Let $f: x \rightarrow x^2 + 3$ be a function on the set of real numbers onto set of real numbers. Find the inverse function $f^{-1}(x)$. (a) $x^2 - 3$ (b) $x^2 + 3$ (c) $\sqrt{x + 3}$ (d) $\sqrt{x - 3}$

12. Find without the use of table the possible value of y if $5^{y^2 - 16} = 625(5^{8y})$.
 (a) 10 and 2 (b) 10 and -2 (c) 10 and 5 (d) -10 and -1

13. If $\frac{1}{8}x = 1.9^2 - 1.1^2$ find the exact value of x . (a) 19.2 (b) 12.1 (c) 29.1 (d) 91.2

14. Find the value of x if $\log_2 x^2 = -8$ (a) $1/16$ (b) 16 (c) -64 (d) 64

15. Given that $\sqrt{5} = 2.236$, evaluate $\frac{3}{2\sqrt{5}}$ (a) 0.7608 (b) 0.6708 (c) 0.8076 (d) 0.7806

16. Simplify $\frac{4!4!}{2!2!2!}$ (a) $2!$ (b) $4!$ (c) $3 \times 4!$ (d) $4 \times 4!$

17. Find the number of ways of arranging the word MATHEMATICS
 (a) 4989600 (b) 4898006 (c) 4889006 (d) 4889600

18. A committee consisting of 3 men and 5 women is selected from 5 men and 10 women. Find how many ways the committee can be formed (a) 262 (b) 226 (c) 622 (d) 2520

1

19. A school Principal and his wife as well as three other teachers are to be seated in a row so that the principal and his wife are next to each other. Find the number of ways this can be done.

- (a) 120 (b) 12 (c) 24 (d) 102

20. Simplify $\sqrt{1000} - \sqrt{40} - \sqrt{90}$. (a) $10\sqrt{5}$ (b) $5\sqrt{10}$ (c) $2\sqrt{10}$ (d) $2\sqrt{5}$

21. Given that x is real and $x \neq 0$, simplify the expression $\left(x - \frac{1}{x}\right)^3 + \left(x + \frac{1}{x}\right)^3$

- (a) $2\left(x^3 + \frac{3}{x}\right)$ (b) $2\left(x^3 + \frac{x}{3}\right)$ (c) $2x^3 + \frac{3}{x}$ (d) $x^3 + \frac{3}{x}$

22. Simplify $\frac{\log_{10} 125}{\log_{10} 25}$. (a) $2/3$ (b) 5 (c) $1/5$ (d) $3/2$

23. Find the value of $\frac{6^{1/2} 9^{1/4}}{216^{1/4}}$ (a) $\left(\frac{2}{3}\right)^{1/4}$ (b) $\left(\frac{3}{4}\right)^{1/4}$ (c) $\left(\frac{3}{2}\right)^{1/4}$ (d) $\left(\frac{1}{3}\right)^{1/4}$

24. Solve $\log_2 x + \log_x 4 = 3$ (a) 2 or 4 (b) 1 or $\frac{1}{4}$ (c) 1 or -2 (d) $\frac{1}{2}$ or -4

25. Solve for x ; $9^{x+1} - 10 \cdot 3^x + 1 = 0$ (a) 1 or 9 (b) 1 or $1/9$ (c) 1 or 0 (d) 0 or -2

26. Express $\frac{3\sqrt{2}-\sqrt{3}}{2\sqrt{3}-\sqrt{2}}$ in the form $\frac{\sqrt{m}}{\sqrt{n}}$, where m and n are whole numbers

- (a) $\frac{\sqrt{6}}{\sqrt{10}}$ (b) $\frac{\sqrt{150}}{\sqrt{100}}$ (c) $\frac{2\sqrt{6}}{\sqrt{10}}$ (d) $\frac{5\sqrt{6}}{\sqrt{10}}$

27. Solve the equation $\sqrt{x+8} + \sqrt{x+1} = 7$ (a) 8 (b) 9 (c) 7 (d) 6

28. The 8th term of a linear sequence is 18 and the 12th term is 26. Find the first term and the common difference (a) 4 and 2 (b) 2 and 4 (c) 3 and 4 (d) 1 and 4

29. If $(y+1)$, $(2y-1)$ and $(3y+1)$ are three consecutive terms of a G.P., find the common ratio.

- (a) 0 and 4 (b) 4 and -8 (c) 8 (d) 0 and 8

30. Evaluate $\sum_{k=1}^{100} (3k+2)$ (a) 31550 (b) 302 (c) 5 (d) 15350

31. If $\frac{16}{9}$, x , 1, y are in GP. Find the product of x and y (a) $\frac{1}{4}$ (b) $\frac{4}{3}$ (c) 1 (d) $-\frac{4}{3}$

32. Find the 8th term of an exponential sequence whose first term is 3 and whose common ratio is 2.

- (a) 128 (b) 384 (c) 4374 (d) 4473

33. Find p in the function $y = px^3 + 2x^2 - 3x + 4$ if y has value zero when $x = 1$

- (a) 0 (b) -3 (c) 2 (d) -2

34. Find the remainder when $x^3 - 2x^2 + 5x + 8$ is divided by $x - 3$

- (a) 32 (b) 27 (c) 24 (d) 42

35. If α and β are the roots of equation $4x^2 - 4x - 3 = 0$. Find $\beta\alpha^3 + \alpha\beta^2$

- (a) $\frac{3}{4}$ (b) $\frac{7}{4}$ (c) $\frac{4}{7}$ (d) $\frac{21}{16}$

36. Resolve into its partial fraction $\frac{7x-1}{(3x-1)(2x+3)}$

- (a) $\frac{4}{3x-1} + \frac{11}{2x+3}$ (b) $\frac{4}{3x-1} - \frac{11}{2x+3}$ (c) $\frac{1}{11} \left(\frac{4}{3x-1} + \frac{23}{2x+3} \right)$ (d) $\frac{1}{11} \left(\frac{4}{3x-1} + \frac{11}{2x+3} \right)$

37. If $\frac{1}{(k+2)(k+4)} = \frac{A}{k+2} + \frac{B}{k+4}$, Find the sum of A and B . (a) 1 (b) $\frac{1}{2}$ (c) $\frac{1}{4}$ (d) 0

38. Solve for x and y ; $\frac{x}{2} + \frac{y}{3} = \frac{-1}{6}$, $x - 2 = -2$, (a) $\frac{3}{4}, \frac{5}{8}$ (b) $\frac{-3}{4}, \frac{5}{8}$ (c) $\frac{3}{4}, \frac{-5}{8}$ (d) $\frac{3}{4}, \frac{-5}{8}$

39. Evaluate $\frac{-2+4i}{1+i}$

- (a) $\frac{2+6i}{3}$ (b) $1+3i$ (c) $\frac{2+6i}{2}$ (d) $1-3i$

40. Evaluate: $\log_5 125 + \log_3 81 - \log_2 64$. (a) 13 (b) 3 (c) -1 (d) 5

SECTION B

1(a) Consider the following relations on the set $A = \{1, 2, 3, 4\}$

- $R_1 = \{(1, 1), (1, 2), (2, 3), (1, 3), (4, 4)\}$
 $R_2 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$
 $R_3 = \{(1, 3), (2, 1)\}$
 $R_4 = \emptyset$
 $R_5 = A \times A$

Determine which of the relation(s) are (i) Reflexive (ii) Symmetric (iii) Transitive.

(b) Prove by Mathematical Induction that:

$$1 + a + a^2 + \dots + a^n = \frac{a^{n+1} - 1}{a - 1}, \quad a \neq 1$$

2(a) If $Z_1 = 2 - i$, $Z_2 = 4i + 2$ and $\frac{1}{Z_3} = +Z_1^{-1} + Z_2^{-1}$, evaluate Z_3 in the form $x + iy$.

(b) Express $-6 - 4i$ in polar form.

UPLOADED BY
ABU JAWAL